Do 7.43 (a-f)

guessing and is designed to discourage test takers from choosing answers at random.

a. It can be shown that if a totally unprepared student answers all 50 questions by just selecting one of the five answers at random, then $\mu_{x_i} = 10$ and $\mu_{x_2} = 40$. What is the mean value of the total score, y? Does this surprise you? Explain.

b. Explain why it is unreasonable to use the formulas given in this section to compute the variance or standard deviation of y.

7.43 Consider a large ferry that can accommodate cars and buses. The toll for cars is \$3, and the toll for buses is \$10. Let x and y denote the number of cars and buses, respectively, carried on a single trip. Cars and buses are accommodated on different levels of the ferry, so the number of buses accommodated on any trip is independent of the number of cars on the trip. Suppose that x and y have the following probability distributions:

x	0	1	2	3	4	5
$p(x)$	.05	.10	.25	.30	.20	.10

y	0	1	2
$p(y)$	.50	.30	.20

a. Compute the mean and standard deviation of x.

b. Compute the mean and standard deviation of y.

c. Compute the mean and variance of the total amount of money collected in tolls from cars.

d. Compute the mean and variance of the total amount of money collected in tolls from buses.

e. Compute the mean and variance of z = total number of vehicles (cars and buses) on the ferry.

f. Compute the mean and variance of w = total amount of money collected in tolls.

7.44 Consider a game in which a red die and a blue die are rolled. Let x_R denote the value showing on the uppermost face of the red die, and define x_B similarly for the blue die.

a. The probability distribution of x_R is:

x_R	1	2	3	4	5	6
$p(x_R)$	1/6	1/6	1/6	1/6	1/6	1/6

Find the mean, variance, and standard deviation of x_R.

b. What are the values of the mean, variance, and standard deviation of x_B? (You should be able to answer this question without doing any additional calculations.)

c. Suppose that you are offered a choice of the following two games:

Game 1: Costs \$7 to play, you win y_1 dollars, where $y_1 = x_R + x_B$.

Game 2: Doesn't cost anything to play initially, but you "win" $3y_2$ dollars, where $y_2 = x_R - x_B$. If y_2 is negative, you must pay that amount; if it is positive, you receive that amount.

For Game 1, the net amount won in a game is $w_1 = y_1 - 7 = x_R + x_B - 7$. What are the mean and standard deviation of w_1?

d. For Game 2, the net amount won in a game is $w_2 = 3y_2 = 3(x_R - x_B)$. What are the mean and standard deviation of w_2?

e. Based on your answers to Parts (c) and (d), if you had to play, which game would you choose and why?

▪ 7.5 Binomial and Geometric Distributions

In this section we introduce two of the more commonly encountered discrete probability distributions: the binomial distribution and the geometric distribution. These distributions arise when the experiment of interest consists of making a sequence of dichotomous observations (two possible values for each observation). The process of making a single such observation is called a *trial*. For example, one characteristic of blood type is Rh factor, which can be either positive or negative. We can think of an experiment that consists of noting the Rh factor for each of 25 blood donors as a sequence of 25 dichotomous trials, where each trial consists of observing the Rh factor (positive or negative) of a single donor.

We could also conduct a different experiment that consists of observing the Rh factor of blood donors until a donor who is Rh-negative is encountered. This second experiment can also be viewed as a sequence of dichotomous trials, but the total number of trials in this experiment is not predetermined, as it was in the previous example, where we knew in advance that there would be 25 trials. Experiments of the two types just described are characteristic of those leading to the binomial and the geometric probability distributions, respectively.

2. A department supervisor is considering purchasing one of two comparable photocopy machines, *A* or *B*. Machine *A* costs $10,000 and machine *B* costs $10,500. This department replaces photocopy machines every three years. The repair contract for machine *A* costs $50 per month and covers an unlimited number of repairs. The repair contract for machine *B* costs $200 per repair. Based on past performance, the distribution of the number of repairs needed over any one-year period for machine *B* is shown below.

Number of Repairs	0	1	2	3
Probability	0.50	0.25	0.15	0.10

You are asked to give a recommendation based on overall cost as to which machine, *A* or *B*, along with its repair contract, should be purchased. What would your recommendation be? Give a statistical justification to support your recommendation.

1. IQ scores are Normally distributed with $\mu = 100$ and $\sigma = 15$. If two people are selected at random, what is the probability that the first person's score will be at least 30 points higher than the second person's score?

2. For an upcoming concert, each customer may purchase up to 3 child tickets and 3 adult tickets. Let C be the number of child tickets purchased by a single customer. The probability distribution of the number of child tickets purchased by a single customer is given in the table below.

c	0	1	2	3
$p(c)$	0.4	0.3	0.2	0.1

(a) Compute the mean and the standard deviation of C.

(b) Suppose the mean and the standard deviation for the number of adult tickets purchased by a single customer are 2 and 1.2, respectively. Assume that the numbers of child tickets and adult tickets purchased are independent random variables. Compute the mean and the standard deviation of the total number of adult and child tickets purchased by a single customer.

(c) Suppose each child ticket costs \$15 and each adult ticket costs \$25. Compute the mean and the standard deviation of the total amount spent per purchase.

3. There are 4 runners on the New High School team. The team is planning to participate in a race in which each runner runs a mile. The team time is the sum of the individual times for the 4 runners. Assume that the individual times of the 4 runners are all independent of each other. The individual times, in minutes, of the runners in similar races are approximately normally distributed with the following means and standard deviations.

	Mean	Standard Deviation
Runner 1	4.9	0.15
Runner 2	4.7	0.16
Runner 3	4.5	0.14
Runner 4	4.8	0.15

(a) Runner 3 thinks that he can run a mile in less than 4.2 minutes in the next race. Is this likely to happen? Explain.

(b) The distribution of possible team times is approximately normal. What are the mean and standard deviation of this distribution?

(c) Suppose the team's best time to date is 18.4 minutes. What is the probability that the team will beat its own best time in the next race?

5. Following are parts of the probability distributions for the random variables X and Y.

x	$P(x)$	y	$P(y)$
1	?	1	.4
2	.2	2	?
3	.3	3	.1
4	?		

If X and Y are independent and the joint probability $P(X = 1, Y = 2) = .1$, what is $P(X = 4)$?

(A) .1
(B) .2
(C) .3
(D) .4
(E) .5

6. Suppose X and Y are random variables with $E(X) = 25$, $\text{var}(X) = 3$, $E(Y) = 30$, and $\text{var}(Y) = 4$. What are the expected value and variance of the random variable $X + Y$?

(A) $E(X + Y) = 55$, $\text{var}(X + Y) = 3.5$
(B) $E(X + Y) = 55$, $\text{var}(X + Y) = 5$
(C) $E(X + Y) = 55$, $\text{var}(X + Y) = 7$
(D) $E(X + Y) = 27.5$, $\text{var}(X + Y) = 7$
(E) There is insufficient information to answer this question.

7. Suppose X and Y are random variables with $\mu_x = 10$, $\sigma_x = 3$, $\mu_y = 15$, and $\sigma_y = 4$. Given that X and Y are independent, what are the mean and standard deviation of the random variable $X + Y$?

(A) $\mu_{x+y} = 25$, $\sigma_{x+y} = 3.5$
(B) $\mu_{x+y} = 25$, $\sigma_{x+y} = 5$
(C) $\mu_{x+y} = 25$, $\sigma_{x+y} = 7$
(D) $\mu_{x+y} = 12.5$, $\sigma_{x+y} = 7$
(E) There is insufficient information to answer this question.

8. Suppose X and Y are random variables with $E(X) = 500$, $\text{var}(X) = 50$, $E(Y) = 400$, and $\text{var}(Y) = 30$. Given that X and Y are independent, what are the expected value and variance of the random variable $X - Y$?

(A) $E(X - Y) = 100$, $\text{var}(X - Y) = 20$
(B) $E(X - Y) = 100$, $\text{var}(X - Y) = 80$
(C) $E(X - Y) = 900$, $\text{var}(X - Y) = 20$
(D) $E(X - Y) = 900$, $\text{var}(X - Y) = 80$
(E) There is insufficient information to answer this question.

9. Suppose the average height of policemen is 71 inches with a standard deviation of 4 inches, while the average for policewomen is 66 inches with a standard deviation of 3 inches. If a committee looks at all ways of pairing up one male with one female officer, what will be the mean and standard deviation for the difference in heights for the set of possible partners?

(A) Mean of 5 inches with a standard deviation of 1 inch
(B) Mean of 5 inches with a standard deviation of 3.5 inches
(C) Mean of 5 inches with a standard deviation of 5 inches
(D) Mean of 68.5 inches with a standard deviation of 1 inch
(E) Mean of 68.5 inches with a standard deviation of 3.5 inches

Name: _____ Date: _____

Show work for all parts.

HSPA math scores for JFK high school are Normally distributed with a mean of 234 and a standard deviation of 21. HSPA math scores for Colonia High School students are Normally distributed with a mean 223 and a standard deviation of 18. A student must score higher than a 200 to pass the HSPA. Assume that scores for students at the two schools are independent.

(a) For each school, calculate the percentage of students who passed the HSPA.

(b) If one student from each school is selected at random, what is the probability that they both pass the HSPA?

(c) If one student from each school is selected at random, what is the probability that at least one of them will fail the HSPA?

(d) What is the probability that a randomly selected JFK student will score at least 25 points higher than a randomly selected Colonia student?

Binomial Probability Problems

Write out the calculator function(s) you used to answer each question.

1. A certain tennis player makes a successful first serve 70% of the time. Assume that each serve is independent of the others. If she serves 6 times, what is the probability that she gets…

a) all 6 serves in?

b) exactly 4 serves in?

c) at least 4 serves in?

d) no more than 4 serves in?

2. The tennis player in problem 1 has 80 first serves in a match. What is the probability that she will get between 52 and 58 serves in? Hint: Do not do binompdf for all numbers between 52 and 58.

3. A wildlife biologist examines frogs for a genetic trait he suspects may be linked to sensitivity to industrial toxins in the environment. Previous research had established that this trait is usually found in 1 of every 8 frogs. He collects and examines a dozen frogs. If the frequency of the trait has not changed, what's the probability that he finds the trait in…

a) none of the 12 frogs?

b) at least 2 frogs?

c) 3 or 4 frogs?

d) no more than 4 frogs?

4. Can a Normal curve be used to model the binomial distribution in problem 1? If yes, use the Normal approximation to calculate the answer to part (c). If not, explain why not.

5. Can a Normal curve be used to model the binomial distribution in problem 2? If yes, use the Normal approximation to calculate the answer. If not, explain why not.

Ch 7 and 8 practice open-ended section:

1. In a game of 4-spot Keno, the player picks 4 numbers from 1-80. The casino randomly selects 20 winning numbers from 1 to 80. The table below shows the possible outcomes of the game and their probabilities, along with the amount of money (payout) that the player wins for a $1 bet.

Matches	0	1	2	3	4
Payout	$0	$0	$1	$3	$120
Probability	.308	.433	.213	.043	.003

Let X = the <u>payout</u> for a <u>single $1 bet</u>.

(a) Calculate the mean and standard deviation of X. What does the expected value tell you about playing this game?

(b) Jerry places a single $5 bet on 4-spot Keno. Find the expected value and the standard deviation of his winnings. (Use your answers to part a).

(c) Marla plays 5 separate games of 4-spot Keno, betting $1 each time. Find the expected value and the standard deviation of her total winnings. (Use your answers to part a).

2. Ed and Andrea attend the same high school, but are in different math classes. The time, E, that it takes Ed to do his math homework follows a Normal distribution with mean 30 min and standard deviation 5 minutes. Andrea's math homework time, A, follows a Normal distribution with mean 50 minutes and standard deviation of 10 minutes. Assume that the times are independent.

(a) Randomly select on math assignment of Ed's and one math assignment of Andrea's. Let the random variable D be the difference in the amount of time each student spent on their assignments. $D = A - E$. Find the mean and standard deviation of D.

(b) Find the probability that Ed spent longer on his assignment than Andrea did on hers.

(c) Find the probability that Ed spent longer on his assignment two nights in a row.

(d) On a given night, Ed and Andrea each have an assignment. What is the probability they will both finish their individual assignments in under 35 minutes?

(e) What is the probability that their combined time for math homework is less than an hour?

3. To demonstrate his pizza making skills, Joe Corbi tosses pepperoni slices onto an individual size pizza from the opposite side of the kitchen. He is able to have the pepperoni land on the pizza with a probability of .85. For each pizza he throws 12 pepperonis. Assume that each throw is independent of the next. Let X = the number of pepperoni slices that land on the pizza.

(a) Does X describe a binomial setting or a geometric setting? Explain how you know.

(b) Calculate the mean and standard deviation for the number of slices that land on the pizza out of 12 throws.

(c) Find the probability that he gets at least 9 slices to land on the pizza.

(d) Can a Normal curve be used to approximate the binomial distribution described in this problem? Explain why or why not.

4. Assume that 16% of the people in the United States have blue eyes. If we select 7 people at random, find the probability for each of the outcomes presented below.

(a) The first blue-eyed person is the fifth person chosen.

(b) There is at least one blue-eyed person in the group.

(c) The first blue-eyed person is the second or third person chosen.

(d) There are more than 2 blue-eyed people in the group.

(e) How many blue-eyed people would you expect to be selected?